Principal Data Engineer

Abbott · United States > Madison : 1 Exact Lane

Job ID (this archive)	HI-036
Employer	Abbott
Location(s)	United States > Madison : 1 Exact Lane
Career stage (archive tag)	Experienced
Location group	Other U.S.
Majors (archive tags)	Computer Science, Data Science
Pay listed	$129,300 – $258,700 per year
Employment type	Full time
Posted / listing date	2026-09-24
Employer requisition	31161904
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://abbott.wd5.myworkdayjobs.com/abbottcareers/job/United-States--Madison--1-Exact-Lane/Principal-Data-Engineer_31161904-2
Captured	2026-10-07T20:10:54.109387 UTC
Course sheet refs	HI-036

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Job description

Abbott is a global healthcare leader that helps people live more fully at all stages of life. Our portfolio of life-changing technologies spans the spectrum of healthcare, with leading businesses and products in diagnostics, medical devices, nutritionals and branded generic medicines. Our 122,000 colleagues serve people in more than 160 countries.

JOB DESCRIPTION:

Position Overview

The Principal Data Engineer (IC) is a senior individual contributor and the accountable technical leader for assigned cross-domain initiatives and enterprise data engineering capabilities. The role owns integrated technical direction and technical outcomes for work spanning multiple data domains, defines and stewards enterprise engineering standards and reference architectures, and drives convergence where duplicated or inconsistent solutions create enterprise cost, risk, or operational burden. The role advises on scope, sequencing, capacity, dependencies, and technical debt, but does not independently commit domain resources or business delivery dates. This position has no people-management responsibility.

Enterprise Data operates a domain-aligned model built on Databricks and Unity Catalog. Working with Domain Leaders, Staff Engineers, Platform Engineering, and partner organizations, the role converts ambiguous enterprise needs into executable architecture and carries the most complex or highest-risk work through validation and production. The role remains hands-on through prototyping, reference implementations, critical-path development, design and code review, and production problem solving.

This role is based in Madison, WI. Relocation assistance may be available for qualified candidates.

Essential Duties

Include, but are not limited to, the following:

Cross-domain technical leadership and delivery

Own the technical outcome of assigned cross-domain initiatives from initial ambiguity through production and lifecycle support.

Convert enterprise needs into integrated architecture and executable technical plans, decomposing complex work into deliverable increments across domains.

Coordinate technical execution across domain Staff Engineers and partner teams, identifying dependencies and architectural risks, and escalating decisions requiring business or delivery authority.

Remain hands-on through prototypes, reference implementations, critical-path development, technical validation, and production problem solving.

Enterprise architecture and standards

Define, steward, and evolve enterprise data engineering standards, patterns, and reference architectures, and drive convergence where inconsistent or duplicative implementations create enterprise cost, risk, or operational burden.

Lead architecture for assigned enterprise capabilities, including semantic and metrics layers, canonical data models, batch, API, event-driven, and streaming patterns, and governed data products supporting analytics, machine learning, and AI.

Establish enterprise data-contract standards covering schemas, service expectations, compatibility, breaking-change policy, and producer-consumer responsibilities, and partner with Platform Engineering to convert recurring cross-domain needs into shared capabilities.

Technical leadership and mentorship

Lead enterprise design and code reviews for high-complexity or cross-domain work, and coach and mentor Staff and Senior Engineers across domains without formal people authority.

Communicate architecture and technical tradeoffs clearly to engineering, business, and executive stakeholders, and build reusable guidance that increases engineering consistency across Enterprise Data.

Operational excellence, security, and efficiency

Evaluate architecture tradeoffs across reliability, scalability, performance, security, privacy, operability, adoption, and technical cost, and drive cost-efficient use of compute, storage, streaming, and orchestration.

Serve as the enterprise technical escalation for incidents involving multiple domains or shared architecture patterns, and lead root-cause analysis and preventive changes for recurring enterprise issues.

Design and review architectures handling protected health information to meet applicable security, privacy, lineage, audit, Quality Management System, HIPAA, CLIA, and regulatory requirements.

Provide technical direction and due diligence for vendor and external-partner solutions, and advance responsible engineering practices including approved AI-assisted development capabilities.

Ability to work nights and/or weekends, as needed.

Minimum Qualifications

Bachelor's Degree in Data Science, Computer Science, Information Systems, Mathematics, or Engineering.

Expert-level experience with software development design and development and with relevant domain specific skills (see below).

Spark on Databricks or comparable platforms; Python, Scala, SQL, and Snowflake experience.

ETL and ELT data pipelines, including batch and event-driven patterns.

Designing and implementing data modeling solutions using relational, dimensional, and/or NoSQL databases.

Database architecture testing methodology, including execution of test plans, debugging, and testing scripts and tools.

Open data file and table formats (Parquet, Avro, Delta Lake); cloud infrastructure and delivery services (AWS, S3, SQS, and GitLab CI/CD).

REST API development; familiarity with BI concepts and Tableau performance considerations.

Agile development tools; including, but not limited to, JIRA, Confluence repository.

Demonstrated ability to lead through influence across multiple teams and communicate complex technical decisions to senior engineering, business, and executive stakeholders.

Demonstrated ability to perform the essential duties of the position with or without accommodation.

Preferred Qualifications

Databricks, Apache Spark, Delta Lake, and Unity Catalog at enterprise scale.

Kafka, change data capture, and production event-streaming architectures.

Semantic or metrics layer design, canonical data models, and governed data products.

Cloud data architecture in AWS, Azure, or Google Cloud Platform.

Life sciences, diagnostics, or clinical laboratory environments involving protected health information, HIPAA, CLIA, FDA, or Quality Management System requirements.

Technical assessment and architecture direction for vendor and external-partner platforms.

The base pay for this position is

$129,300.00 - $258,700.00 In specific locations, the pay range may vary from the range posted.

JOB FAMILY:
Product Development

DIVISION:
ONCO Cancer Diagnostics

LOCATION:
United States > Madison : 1 Exact Lane

ADDITIONAL LOCATIONS:

WORK SHIFT:
Standard

TRAVEL:
Yes, 5 % of the Time

MEDICAL SURVEILLANCE:
No

SIGNIFICANT WORK ACTIVITIES:
Continuous sitting for prolonged periods (more than 2 consecutive hours in an 8 hour day), Keyboard use (greater or equal to 50% of the workday)

Abbott is an Equal Opportunity Employer of Minorities/Women/Individuals with Disabilities/Protected Veterans.

EEO is the Law link - English: http://webstorage.abbott.com/common/External/EEO_English.pdf

EEO is the Law link - Espanol: http://webstorage.abbott.com/common/External/EEO_Spanish.pdf

Notes for ECS 1100 students

How this connects to your major

Data Science: healthcare data analysis, data engineering, statistics, AI or computational biology; Computer Science: healthcare software, embedded systems, automation or secure computing. These are teaching annotations, not an employer guarantee of eligibility.

Use the degree and experience evidence to distinguish a first job from a later career step.

DS: Named discipline appears in source; consult qualifications for eligibility

CS: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Python, SQL, Machine learning, Data visualization, Cloud, Data pipelines/ETL, Medical-device quality, Verification/validation

Course sheet tags

Majors (teaching tags): DS; CS; Career stage (teaching tag): Senior / leadership; Geography: Other US location; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full time.

Source record

Original posting
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